

AN EXPLICIT FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

ABSTRACT. A countable discrete group is MF in the Carrión–Dadarlat–Eckhardt–Shulman sense if it embeds into the unitary group of a norm quotient $\prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$ of matrix algebras. Whether every group is MF has been a longstanding open question; it is the operator-norm case of a family of metric approximation problems whose normalized-Hamming (sofic) case was solved recently by OpenAI and subsequently generalized by Kun and Thom, and whose unnormalized Frobenius case was solved by De Chiffre, Glebsky, Lubotzky and Thom. We construct an explicit finitely presented group E , together with a designated central involution $w \in E$, such that $w \neq 1$ but *every* homomorphism from E to the unitary group of *every* norm quotient of matrix algebras kills w . Consequently E is not MF, and neither its full nor its reduced group C^* -algebra is an MF algebra. The group is built from $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, one HNN letter implementing the proper endomorphism $(v, A) \mapsto (2v, A)$, and one additional involution; the proof combines the Kazhdan projection of Γ with stable finiteness of matrix coronas. The finite-dimensional case of the kill theorem is elementary and uses neither property (T) nor unitarity; we also show, via quasidiagonality of amenable groups, that the same marked presentation with a cyclic base can have a corona representation in which the marked involution survives.

1. INTRODUCTION

Let $(d_n)_{n \in \mathbb{N}}$ be a sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable C^* -algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group E is MF if there is an *injective* group homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q})$$

for some choice of (d_n) . Equivalently, E admits finite-dimensional approximate unitary representations that are approximately injective, with all defects measured in operator norm. This qualification matters: Shulman notes that a strictly stronger group property has also been called MF [Sh, Definition 3 and footnote 1]. Residually finite groups are MF, amenable groups are MF as

a consequence of the Tikuisis–White–Winter theorem [TWW], free products of amenable groups are MF, and a substantial permanence theory exists [Sh]. Whether *every* countable group is MF has remained open; it is the operator-norm member of the family of approximation problems surveyed in [DGLT, BDL], whose unnormalized-Frobenius case was solved negatively in [DGLT]. OpenAI constructed the first nonsofic group [OAI]; Kun and Thom subsequently analyzed its underlying mechanism and obtained nonsofic generalized wreath products [KT]. In the sofic problem the metric is normalized Hamming distance on permutation groups, not rank distance; the three metric conventions are stated together in [Sh, Introduction]. The operator-norm case resisted both methods: the cohomological argument of [DGLT] needs a submultiplicative norm interacting with a trace, while Hamming-metric arguments do not control operator-norm defects.

This paper answers the question negatively, with an explicit finitely presented example.

Theorem A (non-MF group). *Let $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, let $\alpha: \Gamma \rightarrow \Gamma$ be the injective endomorphism $\alpha(v, A) = (2v, A)$, and let $a = (e_1, 1) \in \Gamma$, which does not lie in $\alpha(\Gamma)$. Let E be the finitely presented group obtained from Γ by adjoining a stable letter t with $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$, an involution c commuting with Γ , and by imposing that*

$$w := [tct^{-1}, a(tct^{-1})a^{-1}]$$

is a central involution (Definition 3.2). Then:

- (1) $w \neq 1$ in E ;
- (2) *for every sequence (d_n) and every group homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ into the corresponding norm matrix corona (1), one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C_{\max}^(E)$ nor $C_{\mathrm{red}}^*(E)$ is an MF C^* -algebra.*

A completely explicit presentation of E — eight generators, with every relation displayed — appears as (5) in Definition 3.2.

The proof implements a C^* -algebraic finiteness argument into a group-level kernel theorem. For orientation, it can be organized into the following five steps. The surrounding compression/lamp architecture has close antecedents; the point specific to this paper is its operator-norm realization and the resulting universal marked kernel.

- (1) *Central corner.* If $\Theta(w) \neq 1$, then $e = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with the image of Θ ; cutting by a lift of e replaces $\mathcal{Q}$ by another norm matrix corona on which $\Theta(w) = -1$ (Lemma 6.3). This cut eliminates small-relative-rank phenomena: on the corner, the deviation of w from 1 has full normalized size.
- (2) *Conjugation representation.* Unitary lifts $U_{g,n}$ of the corner representation conjugate the corner coordinates $M_{r_n}(\mathbb{C})$, and exact multiplicativity of Θ in the quotient makes arbitrary coordinate lifts asymptotically multiplicative in operator norm. Thus $g \mapsto (\mathrm{Ad} U_{g,n})_n$ defines an *exact*

homomorphism β into the norm corona of the algebras $B(L^2(M_{r_n}))$ (Lemma 7.1).

- (3) *Kazhdan projection.* Γ has property (T), so its maximal group C^* -algebra contains the Kazhdan projection p_Γ , whose image under every representation is the projection onto the invariant vectors. The HNN relation $t\Gamma t^{-1} = \alpha(\Gamma) \subseteq \Gamma$ forces, in *every* C^* -representation, the order relation $P \leq \beta(t)P\beta(t)^*$ between the images of p_Γ and its t -conjugate (Lemma 7.2).
- (4) *Stable finiteness.* A norm corona of matrix algebras is finite: isometries are unitaries (Lemma 6.2). Hence the two unitarily equivalent projections in step (3) must be equal: $\beta(t)$ *preserves* the Γ -fixed space.
- (5) *The marked word.* The involution c commutes with Γ , so the matrix vector representing $\Theta(c)$ is Γ -fixed; by step (4) so is the vector representing $\Theta(tct^{-1})$; hence $a \in \Gamma$ asymptotically fixes it, and the commutator word w is forced to 1 in normalized Hilbert–Schmidt norm along the chosen ultrafilter — while the corner normalization of step (1) says it equals -1 in operator norm. The two conclusions contradict each other (Section 8).

Nontriviality of w is witnessed by a completely explicit representation of E into a semidirect product of a Clifford group by a linear group over $\mathbb{Z}[1/2]$ (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the kill theorem is elementary and needs neither property (T), nor unitarity, nor centrality of w :

Theorem B (finite-dimensional obstruction). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi: H \rightarrow \text{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

Applied to $H = E$, Theorem B shows that *every* finite-dimensional representation of E over *every* field kills w : finite-dimensional linear representations do not separate the points of E , no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument for one reason: it makes “the projection onto the Γ -fixed space” an element of the corona, uniformly across all coordinates at once.

Second, the fixed-space rigidity in that argument cannot simply be omitted:

Theorem C (cyclic-base comparison). *Let $G_0 = \text{BS}(1, 2) = \langle \gamma_0, t \mid t\gamma_0 t^{-1} = \gamma_0^2 \rangle$, let $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$, let $X_0 = G_0/\Gamma_0$, and let $E_0 = \text{ClLamp}(X_0) \rtimes G_0$ be the corresponding Clifford extension (Construction 4.1), with central sign z_0 .*

Also define the literal marked-presentation analogue

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_0 is an amenable MF group and $z_0 \neq 1$. There is a quotient homomorphism $E_{\text{BS}} \twoheadrightarrow E_0$ sending w_{BS} to z_0 ; consequently the literal analogous presentation has a corona representation in which its marked involution survives. Nevertheless, every finite-dimensional representation of either group kills the corresponding marked element.

The quotient statement makes the comparison at the level actually used by the proof: the marked-word relations and witness mechanism survive when the Kazhdan base is replaced by $\mathbb{Z}$, but the universal corona-kill conclusion does not. Thus some substitute for the Kazhdan-projection step is essential to this mechanism; this is not a claim that every proof of the concrete Theorem A must invoke property (T).

What this paper does not prove. The proof of Theorem A does *not* show that E is nonhyperlinear; equivalently, it does not show that the canonical group trace, or the associated group von Neumann algebra, fails the relevant Connes-embedding property. It also does not touch the question whether hyperlinear groups are sofic. The finiteness step (4) is special to operator-norm coronas: in a tracial (Hilbert–Schmidt) matrix ultraproduct the conjugation representation acts on the infinite Hilbert space L^2 of the ultraproduct, the relevant fixed-space projections live in a properly infinite algebra, and a projection there can be properly contained in a unitarily equivalent one — the unilateral-shift phenomenon. Section 10 makes this precise. The same distinction explains why the group $E/\langle w \rangle$ and the ambient groups G remain untouched by our argument.

External inputs. The proof of Theorem A uses two non-elementary external inputs beyond elementary linear algebra and C^* -basics: property (T) of $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ (Fact 2.1) and the existence and defining property of the Kazhdan projection (Fact 2.2). Theorem C additionally uses quasidiagonality of amenable group C^* -algebras [TWW] (Fact 2.3). Theorem B uses no external input.

Relation to prior work. The closest conceptual analogue is the recent work on nonsofic groups. OpenAI’s first example and Kun–Thom’s analysis use one-sided containments $g\Gamma g^{-1} \leq \Gamma$, property (T), normalization of fixed or centralizer structures, and generalized lamps in a closely related architecture [OAI, KT]. We therefore make no novelty claim for the compression/lamp paradigm itself. Their rigidity takes place in permutation models with normalized Hamming distance. Here the decisive operator-norm step is instead

$$P \leq \beta(t)P\beta(t)^* \implies P = \beta(t)P\beta(t)^*,$$

because the conjugation representation lands in a stably finite norm matrix corona. The central corner then converts this equality into a fixed nontrivial element killed by every corona representation.

There is also an older analytic ancestor. The Ozawa–Thom argument recorded as [Dad, Proposition 3.19] turns operator-norm quasidiagonal finite-rank projections into almost invariant normalized Hilbert–Schmidt vectors for a conjugation representation and invokes property (T) to obtain an invariant vector. Its conclusion is that an infinite weakly quasidiagonal property-(T) group has an infinite residually finite quotient; it does not produce a universal marked kernel. It is nevertheless direct prior art for the passage from operator-norm control to conjugation on Hilbert–Schmidt operators and property-(T) fixed vectors.

The Kazhdan-projection compression observation needed here is proved directly in Lemmas 7.2 and 7.3; we do not attribute that observation to Carrión–Dadarlat–Eckhardt. Their work is used for the group-MF framework and for comparison examples. In particular, their treatment of Abels’ finitely presented solvable group already supplies an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. Since a finitely generated maximally almost periodic group is residually finite by Mal’cev’s theorem, the broad phenomenon “finitely presented amenable MF but non-MAP” is therefore not new. Theorem C instead gives a calibrated statement about the same marked-sign compression pattern used in Theorem A.

Finally, non-MF-ness of one group C^* -algebra alone does not obstruct the *group* from embedding into a corona, since group MF-ness only requires some quotient of $C_{\max}^*$ to be MF. The contribution asserted here is narrower: stable-finite Kazhdan fixed-space rigidity in a norm matrix corona, together with the central-corner/marked-word device producing a specific nontrivial element in the kernel of every such representation. For context on the approximation landscape see [DGLT, BDL, OAI, KT, Sh, Dad].

Organization. Section 2 fixes notation and states the external inputs. Section 3 constructs E . Section 4 proves $w \neq 1$. Section 5 proves Theorem B. Sections 6–8 prove Theorem A. Section 9 proves Theorem C. Section 10 discusses limitations.

2. NOTATION AND EXTERNAL INPUTS

All groups are discrete and countable. All C^* -homomorphisms and C^* -algebra embeddings are understood to be unital unless explicitly stated otherwise. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. We will use three elementary facts without further comment: $\|x\|_2 \leq \|x\|$; $|\mathrm{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v .

Given a sequence $(d_n)_{n \in J}$ of positive integers indexed by an infinite set $J \subseteq \mathbb{N}$, we write

$$\mathcal{Q}((d_n)_{n \in J}) = \prod_{n \in J}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in J}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded C^* -product by the ideal of sequences with $\|x_n\| \rightarrow 0$. We call any such algebra a *norm matrix corona*. A *corona representation* of a group E is a group homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$ for some norm matrix corona $\mathcal{Q}$; the group E is *MF* if it admits an injective corona representation.

For a group H , $C_{\max}^*(H)$ denotes the maximal group C^* -algebra and $u_h \in C_{\max}^*(H)$ the canonical unitaries. Every unitary representation π of H on a Hilbert space extends uniquely to a $*$ -representation of $C_{\max}^*(H)$, still denoted π . Every group homomorphism $\varphi: H_1 \rightarrow H_2$ induces a unital $*$ -homomorphism $C_{\max}^*(H_1) \rightarrow C_{\max}^*(H_2)$, $u_h \mapsto u_{\varphi(h)}$.

We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

The following two classical facts are the only external inputs to Theorem A.

Fact 2.1 (property (T); [BHV, Sections 1.4–1.5 and 4.2]). *$\mathrm{SL}_3(\mathbb{Z})$ has property (T), the pair $(\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \mathbb{Z}^3)$ has relative property (T), and consequently $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ has property (T). Γ is finitely presented.*

Fact 2.2 (Kazhdan projection; [AW], [BHV, Chapter 6]). *Let Γ be a group with property (T). There is a projection $p_\Gamma \in C_{\max}^*(\Gamma)$ such that for every unitary representation π of Γ on a Hilbert space $\mathcal{H}$, the operator $\pi(p_\Gamma)$ is the orthogonal projection onto the subspace $\mathrm{Fix}(\pi) = \{\xi \in \mathcal{H} : \pi(\gamma)\xi = \xi \text{ for all } \gamma \in \Gamma\}$. Moreover $u_\gamma p_\Gamma = p_\Gamma u_\gamma = p_\Gamma$ for every $\gamma \in \Gamma$.*

The identities follow from the defining property: the left identity holds in every representation, and the right identity follows by taking adjoints (or replacing γ by γ^{-1}); $C_{\max}^*(\Gamma)$ has a faithful representation.

The only additional input, used exclusively in Section 9, is:

Fact 2.3 ([TWW, Corollary B]). *The reduced group C^* -algebra of a countable amenable group is quasidiagonal.*

Fact 2.4 (quasidiagonal embedding; see [BO, Chapter 7]). *Every separable unital quasidiagonal C^* -algebra embeds unitaly into a norm matrix corona. Consequently every countable amenable group is MF.*

3. THE GROUP

Throughout, $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ with multiplication $(v, A)(w, B) = (v + Aw, AB)$.

We first record concrete presentation data. By [CRW, Theorem 2], $\mathrm{SL}_3(\mathbb{Z})$ has the presentation

$$(2) \quad \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

where the generators are the matrices

$$x = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}.$$

Let $v_i = (e_i, 1)$ for $1 \leq i \leq 3$. Reading off the columns of these matrices gives the following complete finite presentation of the semidirect product:

$$(3) \quad \Gamma = \left\langle v_1, v_2, v_3, x, y, z \left| \begin{array}{l} \text{the eight relations in (2),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_1, \quad xv_2x^{-1} = v_2^{-1}v_3, \quad xv_3x^{-1} = v_1v_2^{-1}, \\ yv_1y^{-1} = v_3, \quad yv_2y^{-1} = v_1, \quad yv_3y^{-1} = v_2, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right. \right\rangle.$$

Lemma 3.1. *The map $\alpha(v, A) = (2v, A)$ is an injective endomorphism of Γ with image $2\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ of index 8, and $a := (e_1, 1) \notin \alpha(\Gamma)$.*

Proof. We compute

$$\begin{aligned} \alpha((v, A)(w, B)) &= \alpha(v + Aw, AB) = (2v + 2Aw, AB) \\ &= (2v, A)(2w, B) = \alpha(v, A)\alpha(w, B), \end{aligned}$$

so α is a homomorphism, clearly injective, with image $2\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$; the index equals $[\mathbb{Z}^3 : 2\mathbb{Z}^3] = 8$. If $(e_1, 1) = (2v, A)$ then $A = 1$ and $e_1 = 2v$, which is impossible in $\mathbb{Z}^3$. $\square$

Let G denote the ascending HNN extension

$$(4) \quad G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \ (\gamma \in \Gamma) \rangle.$$

We will use G only as an auxiliary group receiving no injectivity claims; every statement we need about it is either a presentation identity or is verified inside an explicit linear quotient (Lemma 4.2).

Definition 3.2. Use the six base generators and the relations displayed in (3), and put $a = v_1$. Define

$$(5) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{all relations in (3),} \\ tv_it^{-1} = v_i^2 \ (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \quad w^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$.

Thus (5) is an explicit presentation with eight generators and a fully displayed finite list of relators. The presentation identities give a canonical homomorphism $G \rightarrow E$ and a canonical (not a priori injective) homomorphism $\Gamma \rightarrow E$; we write γ, t, c for the images in E . In E the element w is a central involution by fiat, and the relations $t\gamma t^{-1} = \alpha(\gamma)$, $[c, \gamma] = 1$ hold for all $\gamma \in \Gamma$ because they hold on generators.

4. THE WITNESS: $w \neq 1$

Construction 4.1 (Clifford groups). Let X be a set. Define $\text{Cl}(X)$ to be the quotient of the free unital associative real algebra on symbols $(c_x)_{x \in X}$ by the relations

$$c_x^2 = 1, \quad c_x c_y = -c_y c_x \quad (x \neq y).$$

The ordered square-free monomials form the usual basis (for arbitrary X , each monomial has finite support), so the algebra is nonzero and $-1 \neq 1$. Let $\text{ClLamp}(X) \leq \text{Cl}(X)^\times$ be the subgroup generated by $\{\pm 1\} \cup \{c_x : x \in X\}$ and write $z = -1 \in \text{ClLamp}(X)$, a central involution. Every permutation of X induces an algebra automorphism of $\text{Cl}(X)$ mapping $c_x \mapsto c_{\sigma(x)}$ and fixing z ; this gives an action of any group acting on X by automorphisms of $\text{ClLamp}(X)$, and a semidirect product $\text{ClLamp}(X) \rtimes G$ for any G -set X .

Lemma 4.2 (linear model). *Embed $\Gamma \hookrightarrow \text{GL}_4(\mathbb{Z})$ by $(v, A) \mapsto \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}$ and let $D = \text{diag}(2, 2, 2, 1) \in \text{GL}_4(\mathbb{Z}[1/2])$. Then $D\gamma D^{-1} = \alpha(\gamma)$ for all $\gamma \in \Gamma$, so $\bar{G} := \langle \Gamma, D \rangle \leq \text{GL}_4(\mathbb{Z}[1/2])$ is a quotient of G via $t \mapsto D$. In the $\bar{G}$ -set $\bar{X} := \bar{G}/\Gamma$ the cosets $D\Gamma$ and $aD\Gamma$ are distinct.*

Proof. $D \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D^{-1} = \begin{pmatrix} A & 2v \\ 0 & 1 \end{pmatrix}$, which is the matrix of $\alpha(v, A)$; so the assignment $\gamma \mapsto \gamma$, $t \mapsto D$ respects the relations (4). If $aD\Gamma = D\Gamma$ then $D^{-1}aD \in \Gamma$ inside $\text{GL}_4(\mathbb{Z}[1/2])$. But $D^{-1}aD = \begin{pmatrix} 1 & \frac{1}{2}e_1 \\ 0 & 1 \end{pmatrix}$ has a non-integral entry, so it does not lie in the integral group Γ . $\square$

Proposition 4.3. $w \neq 1$ in E .

Proof. Let $\bar{X} = \bar{G}/\Gamma$ as in Lemma 4.2 and form $W := \text{ClLamp}(\bar{X}) \rtimes \bar{G}$ as in Construction 4.1. Define a map on the generators of (5) by

$$\begin{aligned} g &\mapsto g \in \bar{G} \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto D \in \bar{G}, \quad c \mapsto c_o, \quad o := \Gamma \in \bar{X}. \end{aligned}$$

We check the relations. The base relations in (3) and $tgt^{-1} = \alpha(g)$ for the six base generators hold in $\bar{G}$ by Lemma 4.2. We have $c_o^2 = 1$. For $\gamma \in \Gamma$ we have $\gamma o = o$, so $\gamma c_o \gamma^{-1} = c_{\gamma o} = c_o$, giving the six commutation relations for c . Under this assignment,

$$tct^{-1} \mapsto c_{D\Gamma}, \quad a(tct^{-1})a^{-1} \mapsto c_{aD\Gamma},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{D\Gamma}, c_{aD\Gamma}] = c_{D\Gamma} c_{aD\Gamma} c_{D\Gamma}^{-1} c_{aD\Gamma}^{-1} = -1 = z.$$

Since z is a central involution of W , the remaining relations $w^2 = 1$ and $[w, s] = [w, t] = [w, c] = 1$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $z \neq 1$, and therefore $w \neq 1$ in E . $\square$

Remark 4.4. The target uses the infinite group $\text{CLamp}(\bar{X})$ when $\bar{X}$ is infinite, but each group element involves only finitely many Clifford sites; the remaining data are matrices over $\mathbb{Z}[1/2]$. No normal-form theory for HNN extensions enters. In particular the consistency of the presentation (5) is verified by an explicit linear-algebraic model.

5. THE FINITE-DIMENSIONAL OBSTRUCTION

We prove Theorem B. Everything happens inside the finite-dimensional algebra $\text{End}_k(V)$.

Proof of Theorem B. Let $\pi: H \rightarrow \text{GL}(V)$ be a homomorphism and let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\text{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\text{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(6) \quad C = \Phi(C) = uCu^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ . By (6), $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$, and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

Corollary 5.1. *The group E of Definition 3.2 admits no injective homomorphism into $\text{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, finite-dimensional linear representations do not separate its points: they all identify w with 1. In particular E is not maximally almost periodic or residually finite, $C_{\max}^*(E)$ is not residually finite-dimensional as a C^* -algebra, and no finite quotient of E can witness $w \neq 1$.*

Proof. Apply Theorem B with $H = E$ and the images of Γ , t , c , a in E ; the hypotheses hold by the defining relations (5). Every finite-dimensional representation kills w , and $w \neq 1$ by Proposition 4.3. A finite quotient gives a finite-dimensional representation after composition with a faithful permutation representation, so it also kills w . $\square$

Remark 5.2. The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. The corona of (1) is not finite-dimensional, and the commutant of the image of Γ there is not a coordinatewise object, so the argument does not transfer as stated. Property (T) repairs exactly this: the Kazhdan projection turns “the Γ -fixed subspace” into a single element of the corona, to which the finiteness of the corona (Lemma 6.2) can be applied. The rest of the paper carries this out.

6. NORM MATRIX CORONAS

Throughout this section $\mathcal{Q} = \mathcal{Q}((d_n)_{n \in J})$ is a norm matrix corona, $q: \prod_{n \in J} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ the quotient map.

Lemma 6.1 (lifting).

- (1) Every projection $e \in \mathcal{Q}$ lifts to a sequence $(e_n)_{n \in J}$ of projections, $e_n \in M_{d_n}(\mathbb{C})$.
- (2) Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.

Proof. (1) Lift e to a bounded sequence (x_n) ; replacing x_n by $\frac{1}{2}(x_n + x_n^*)$ and then by the functional-calculus cutoff $f(x_n)$, where $f(t) = \min(1, \max(0, t))$, we may assume $0 \leq x_n \leq 1$. These replacements preserve the quotient class because functional calculus commutes with the quotient map and $f(e) = e$. Since $q((x_n))$ is a projection, $\|x_n^2 - x_n\| \rightarrow 0$, so the spectrum of x_n is eventually contained in $[0, \frac{1}{3}) \cup (\frac{2}{3}, 1]$; for such n put $e_n = \chi_{(\frac{1}{2}, \infty)}(x_n)$, a projection with $\|e_n - x_n\| \rightarrow 0$, and set $e_n = 0$ for the finitely many remaining n . Then (e_n) lifts e .

(2) Lift u to an arbitrary bounded sequence (x_n) . Since $q((x_n))$ is unitary, $\|x_n^* x_n - 1\| \rightarrow 0$, so x_n is eventually invertible; for such n put $u_n = x_n(x_n^* x_n)^{-1/2}$, a unitary with $\|u_n - x_n\| \rightarrow 0$; set $u_n = 1$ for the remaining n . $\square$

Lemma 6.2 (finiteness). *Let $\mathcal{Q}$ be a norm matrix corona.*

- (1) Every isometry in $\mathcal{Q}$ is a unitary; the same holds in every matrix amplification $M_k(\mathcal{Q})$.
- (2) If $p, q \in \mathcal{Q}$ are projections with $p \leq q$ and $v^* v = q$, $vv^* = p$ for some $v \in \mathcal{Q}$, then $p = q$.

Proof. (1) Let $v \in \mathcal{Q}$ with $v^* v = 1$ and lift v to (x_n) with $\|x_n^* x_n - 1\| \rightarrow 0$. Eventually x_n is invertible (a left-invertible square matrix is invertible), and the unitary $u_n = x_n(x_n^* x_n)^{-1/2}$ satisfies $\|u_n - x_n\| \rightarrow 0$; hence $v = q((u_n))$ is unitary. For $M_k(\mathcal{Q})$ note $M_k(\mathcal{Q}) \cong \mathcal{Q}((k d_n)_{n \in J})$, since matrix amplification commutes with products and with the ideal of null sequences.

(2) This is the standard fact that in a finite algebra a projection is not equivalent to a proper subprojection; we include the short verification for completeness. As in any C^* -algebra, $v^* v = q$ gives $v = vq$ (because $\|v(1 - q)\|^2 = \|(1 - q)v^* v(1 - q)\| = \|(1 - q)q(1 - q)\| = 0$) and $vv^* = p$

gives $v = pv$ (because $\|(1-p)v\|^2 = \|(1-p)vv^*(1-p)\| = 0$). From $p \leq q$ we get $qv = q(pv) = pv = v$, hence $(1-q)v = 0$ and, taking adjoints, $v^*(1-q) = 0$ and $(1-q)v^* = (v(1-q))^* = 0$. Then $s := v + (1-q)$ satisfies

$$s^*s = v^*v + v^*(1-q) + (1-q)v + (1-q) = q + 0 + 0 + (1-q) = 1,$$

so s is an isometry, and

$$ss^* = vv^* + v(1-q) + (1-q)v^* + (1-q) = p + 0 + 0 + (1-q) = p + (1-q).$$

By part (1), $ss^* = 1$, i.e. $p + (1-q) = 1$, i.e. $p = q$. $\square$

Lemma 6.3 (central corners). *Let $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ be a corona representation and suppose $\Theta(w) \neq 1$. Then there are an infinite set $J' \subseteq J$, positive integers $(r_n)_{n \in J'}$ with $r_n \leq d_n$, and a corona representation*

$$\Theta_-: E \longrightarrow \mathcal{U}(\mathcal{Q}'), \quad \mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'}),$$

with $\Theta_-(w) = -1$.

Proof. Since $w^2 = 1$ and w is central in E , the element $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(E)$, and $e := \frac{1}{2}(1 - \Theta(w))$ is a projection commuting with $\Theta(E)$, nonzero because $\Theta(w) \neq 1$. By Lemma 6.1(1) lift e to projections (e_n) ; let $J' = \{n \in J : e_n \neq 0\}$ and $r_n = \text{rank } e_n$ for $n \in J'$. The set J' is infinite, since otherwise (e_n) would be a null sequence and $e = 0$.

Consider the map

$$\Psi: \prod_{n \in J'} e_n M_{d_n}(\mathbb{C}) e_n \longrightarrow e \mathcal{Q} e, \quad (x_n)_{n \in J'} \longmapsto q((\tilde{x}_n)_{n \in J}),$$

where $\tilde{x}_n = x_n$ for $n \in J'$ and $\tilde{x}_n = 0$ otherwise. Ψ is a $*$ -homomorphism onto $e \mathcal{Q} e$: surjectivity holds because any $y \in e \mathcal{Q} e$ lifts to (y_n) with $\|e_n y_n e_n - y_n\| \rightarrow 0$, and $e_n y_n e_n = 0$ off J' . Its kernel is exactly the sequences with $\|x_n\| \rightarrow 0$. Choosing unitary identifications $e_n \mathbb{C}^{d_n} \cong \mathbb{C}^{r_n}$ identifies $e_n M_{d_n}(\mathbb{C}) e_n$ with $M_{r_n}(\mathbb{C})$ compatibly with adjoints and norms, so

$$e \mathcal{Q} e \cong \mathcal{Q}((r_n)_{n \in J'}) = \mathcal{Q}',$$

a unital isomorphism carrying e to 1. Finally, since e commutes with $\Theta(E)$, the map $\Theta_-(g) := \Theta(g)e$ is a unital group homomorphism $E \rightarrow \mathcal{U}(e \mathcal{Q} e) = \mathcal{U}(\mathcal{Q}')$, and $\Theta_-(w) = \Theta(w)e = (1 - 2e)e = -e = -1$ in $\mathcal{Q}'$. $\square$

7. THE CONJUGATION REPRESENTATION AND THE KAZHDAN PROJECTION

Fix now a corona representation $\Theta_-: E \rightarrow \mathcal{U}(\mathcal{Q}')$ with $\Theta_-(w) = -1$, where $\mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'})$ as produced by Lemma 6.3. Using Lemma 6.1(2), choose for every $g \in E$ a sequence of unitaries $U_{g,n} \in M_{r_n}(\mathbb{C})$ lifting $\Theta_-(g)$, with $U_{1,n} = 1$.

Let $K_n = L^2(M_{r_n}(\mathbb{C}), \text{tr}_{r_n})$, a Hilbert space of dimension r_n^2 whose bounded operators $B(K_n)$ form the matrix algebra $M_{r_n^2}(\mathbb{C})$, and let

$$\mathcal{B} = \prod_{n \in J'} B(K_n) / \bigoplus_{n \in J'} B(K_n),$$

again a norm matrix corona. For a unitary $U \in M_{r_n}(\mathbb{C})$ let $\text{Ad } U \in \mathcal{U}(B(K_n))$ denote $x \mapsto UxU^*$.

Lemma 7.1.

(1) For unitaries $U, V \in M_r(\mathbb{C})$ and every $x \in M_r(\mathbb{C})$,

$$\|UxU^* - VxV^*\|_2 \leq 2 \|U - V\| \|x\|_2,$$

hence $\|\text{Ad } U - \text{Ad } V\| \leq 2 \|U - V\|$ as operators on the Hilbert space K_n (and therefore as elements of $B(K_n)$).

(2) The class $\beta(g)$ of the sequence $(\text{Ad } U_{g,n})_{n \in J'}$ in $\mathcal{B}$ defines a group homomorphism $\beta: E \rightarrow \mathcal{U}(\mathcal{B})$, independent of the choice of unitary lifts.

(3) β extends to a unital $*$ -homomorphism $\tilde{\beta}: C_{\max}^*(E) \rightarrow \mathcal{B}$ with $\tilde{\beta}(u_g) = \beta(g)$.

Proof. (1) Since $UxU^* - VxV^* = (U - V)xU^* + Vx(U - V)^*$, the ideal property of the normalized Hilbert–Schmidt norm gives

$$\begin{aligned} \|UxU^* - VxV^*\|_2 &\leq \|(U - V)xU^*\|_2 + \|Vx(U - V)^*\|_2 \\ &\leq 2 \|U - V\| \|x\|_2. \end{aligned}$$

Taking the supremum over $\|x\|_2 = 1$ proves the asserted operator-norm bound on K_n .

(2) If $(U_{g,n})$ and $(U'_{g,n})$ both lift $\Theta_-(g)$ then $\|U_{g,n} - U'_{g,n}\| \rightarrow 0$, so by (1) the two sequences of adjoint maps agree in $\mathcal{B}$; this is independence of lifts. For multiplicativity, $U_{g,n}U_{h,n}$ and $U_{gh,n}$ both lift $\Theta_-(gh)$, so $\|U_{g,n}U_{h,n} - U_{gh,n}\| \rightarrow 0$ and, by (1) again, $\beta(g)\beta(h) = \beta(gh)$. Finally $\beta(1) = 1$ since $U_{1,n} = 1$, so each $\beta(g)$ is a unitary with inverse $\beta(g^{-1})$.

(3) Represent $\mathcal{B}$ faithfully and unitaly on a Hilbert space $\mathcal{H}_{\mathcal{B}}$. Then $g \mapsto \beta(g)$ is a unitary representation of E on $\mathcal{H}_{\mathcal{B}}$, which extends to a $*$ -representation $\tilde{\beta}$ of $C_{\max}^*(E)$ on $\mathcal{H}_{\mathcal{B}}$. For $x \in C_{\max}^*(E)$ the operator $\tilde{\beta}(x)$ lies in the norm closure of the linear span of $\{\beta(g) : g \in E\}$, which is contained in the C^* -subalgebra $\mathcal{B}$. $\square$

Recall the auxiliary HNN extension G of (4) and the canonical homomorphisms $\Gamma \rightarrow G \rightarrow E$. Composing, $\tilde{\beta}$ restricts to unital $*$ -homomorphisms of $C_{\max}^*(G)$ and $C_{\max}^*(\Gamma)$ into $\mathcal{B}$; we denote all of them $\tilde{\beta}$, and we write

$$P := \tilde{\beta}(p_{\Gamma}) \in \mathcal{B},$$

a projection.

Lemma 7.2 (order identity). *Let $\iota: C_{\max}^*(\Gamma) \rightarrow C_{\max}^*(G)$ be induced by $\Gamma \rightarrow G$, and write $\bar{p} = \iota(p_{\Gamma})$. Then, in $C_{\max}^*(G)$:*

- (1) $u_{\gamma} \bar{p} = \bar{p}$ for all $\gamma \in \Gamma$;
- (2) $\bar{p}(u_t \bar{p} u_t^*) = \bar{p}$; that is, $\bar{p} \leq u_t \bar{p} u_t^*$.

Consequently, in $\mathcal{B}$: $\beta(\gamma)P = P$ for all $\gamma \in \Gamma$, and $P \leq \beta(t)P\beta(t)^*$.

Proof. Both identities may be checked in an arbitrary unitary representation π of G on a Hilbert space $\mathcal{H}$, since $C_{\max}^*(G)$ admits a faithful one. The composition $\pi \circ \iota$ is the integrated form of $\pi|_{\Gamma}$ (both are $*$ -homomorphisms extending the same unitary representation of Γ), so by Fact 2.2, $\pi(\bar{p})$ is the orthogonal projection onto $\text{Fix}(\pi|_{\Gamma})$, and (1) holds.

For (2): $\pi(u_t \bar{p} u_t^*) = \pi(t) \pi(\bar{p}) \pi(t)^*$ is the orthogonal projection onto $\pi(t) \text{Fix}(\pi|_{\Gamma})$. We claim $\text{Fix}(\pi|_{\Gamma}) \subseteq \pi(t) \text{Fix}(\pi|_{\Gamma})$. Let $\xi \in \text{Fix}(\pi|_{\Gamma})$ and put $\eta = \pi(t)^{-1} \xi$. For $\gamma \in \Gamma$, using $t \gamma t^{-1} = \alpha(\gamma) \in \Gamma$,

$$\pi(\gamma) \eta = \pi(t)^{-1} \pi(t \gamma t^{-1}) \xi = \pi(t)^{-1} \pi(\alpha(\gamma)) \xi = \pi(t)^{-1} \xi = \eta,$$

so $\eta \in \text{Fix}(\pi|_{\Gamma})$ and $\xi = \pi(t) \eta$. The claimed inclusion of ranges gives $\pi(\bar{p}) \leq \pi(u_t \bar{p} u_t^*)$, i.e. (2) in the representation π ; as π was arbitrary and some π is faithful, (2) holds in $C_{\max}^*(G)$. The final statement follows by applying the $*$ -homomorphism $\tilde{\beta}$ (through $C_{\max}^*(G)$). $\square$

Lemma 7.3 (compression equality). $P = \beta(t) P \beta(t)^*$ in $\mathcal{B}$.

Proof. Put $Q_0 = \beta(t) P \beta(t)^*$ and $v = \beta(t) P$. Then $v^* v = P \beta(t)^* \beta(t) P = P$ and $vv^* = Q_0$, and $P \leq Q_0$ by Lemma 7.2. Apply Lemma 6.2(2) with $p := P$, $q := Q_0$, and the element v^* , which satisfies $(v^*)^*(v^*) = vv^* = Q_0$ and $(v^*)(v^*)^* = v^* v = P$; it yields $P = Q_0$. $\square$

We now pass from the corona projection P to its literal fixed-space action on a Hilbert-space ultraproduct. Fix a free ultrafilter ω on the infinite index set J' , and let

$$K_{\omega} = \left\{ (\xi_n)_{n \in J'} : \xi_n \in K_n, \sup_n \|\xi_n\| < \infty \right\} / \left\{ (\xi_n) : \lim_{n \rightarrow \omega} \|\xi_n\| = 0 \right\},$$

the Hilbert-space ultraproduct with inner product $\langle [\xi], [\eta] \rangle = \lim_{n \rightarrow \omega} \langle \xi_n, \eta_n \rangle$.

Lemma 7.4. For a bounded sequence $(T_n)_{n \in J'}$ of operators with class $[T] \in \mathcal{B}$, the assignment $\sigma([T])[(\xi_n)] = [(T_n \xi_n)]$ is a well-defined unital $*$ -homomorphism $\sigma: \mathcal{B} \rightarrow B(K_{\omega})$. Moreover $\sigma(P)$ is the orthogonal projection onto

$$\text{Fix}_{\omega} = \{ \zeta \in K_{\omega} : \sigma(\beta(\gamma)) \zeta = \zeta \text{ for all } \gamma \in \Gamma \}.$$

Proof. If (T_n) is a null sequence in operator norm then $\|T_n \xi_n\| \leq \|T_n\| \sup \|\xi_n\| \rightarrow 0$, in particular along ω ; so σ is well defined on $\mathcal{B}$, and it is clearly a unital $*$ -homomorphism. The composition $\sigma \circ \tilde{\beta}: C_{\max}^*(\Gamma) \rightarrow B(K_{\omega})$ is a $*$ -representation on a Hilbert space extending the unitary representation $\gamma \mapsto \sigma(\beta(\gamma))$; by uniqueness of integrated forms and Fact 2.2, $\sigma(P) = \sigma(\tilde{\beta}(p_{\Gamma}))$ is the orthogonal projection onto the fixed space of that representation, which is Fix_{ω} . $\square$

8. PROOF OF THEOREM A

Proof of Theorem A. Part (1) is Proposition 4.3. For part (2), let $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ be a corona representation and suppose, for contradiction, that $\Theta(w) \neq 1$. Apply Lemma 6.3 to obtain Θ_- with $\Theta_-(w) = -1$ on the corner corona

$\mathcal{Q}'$, and adopt all notation of Section 7: the unitary lifts $U_{g,n}$, the corona $\mathcal{B}$, the homomorphism β , the projection P , the ultraproduct K_ω and the representation σ .

Step 1: the lamp vector is fixed. Each $U_{c,n}$ is a unitary matrix, hence a vector in K_n of norm $\|U_{c,n}\|_2 = 1$; let $\xi_c = [(U_{c,n})] \in K_\omega$, a unit vector. For $\gamma \in \Gamma$ we have $\gamma c \gamma^{-1} = c$ in E , so $U_{\gamma,n} U_{c,n} U_{\gamma,n}^*$ and $U_{c,n}$ both lift $\Theta_-(c)$, whence $\|U_{\gamma,n} U_{c,n} U_{\gamma,n}^* - U_{c,n}\| \rightarrow 0$ and a fortiori the K_n -norms tend to 0. Therefore $\sigma(\beta(\gamma))\xi_c = \xi_c$ for every $\gamma \in \Gamma$, i.e. $\xi_c \in \text{Fix}_\omega$, and by Lemma 7.4, $\sigma(P)\xi_c = \xi_c$.

Step 2: the translated lamp vector is fixed. Let $\xi_d := \sigma(\beta(t))\xi_c = [(U_{t,n} U_{c,n} U_{t,n}^*)]$. By Lemma 7.3 and Step 1,

$$\sigma(P)\xi_d = \sigma(\beta(t)P\beta(t)^*)\sigma(\beta(t))\xi_c = \sigma(\beta(t))\sigma(P)\xi_c = \xi_d,$$

so $\xi_d \in \text{Fix}_\omega$. In particular, since $a \in \Gamma$,

$$(7) \quad \lim_{n \rightarrow \omega} \|U_{a,n} D_n U_{a,n}^* - D_n\|_2 = 0, \quad D_n := U_{t,n} U_{c,n} U_{t,n}^*.$$

Step 3: the commutator word collapses in Hilbert–Schmidt norm. Put $E_n := U_{a,n} D_n U_{a,n}^*$ and $W_n := D_n E_n D_n^{-1} E_n^{-1}$, unitaries in $M_{r_n}(\mathbb{C})$. For unitaries D, E ,

$$\begin{aligned} \|DED^{-1}E^{-1} - 1\|_2 &= \|DE - ED\|_2 \\ &\leq \|(D - E)E\|_2 + \|E(E - D)\|_2 = 2\|D - E\|_2, \end{aligned}$$

using unitary invariance of $\|\cdot\|_2$. With (7),

$$(8) \quad \lim_{n \rightarrow \omega} \|W_n - 1\|_2 = 0.$$

Step 4: the corner normalization forbids the collapse. The finite product W_n of the lifts $U_{t,n}, U_{c,n}, U_{a,n}$ and their adjoints is an operator-norm lift of

$$\Theta_-((tct^{-1}) \cdot a(tct^{-1})a^{-1} \cdot (tct^{-1})^{-1} \cdot (a(tct^{-1})a^{-1})^{-1}) = \Theta_-(w) = -1,$$

so $\|W_n + 1\| \rightarrow 0$ as $n \rightarrow \infty$ in J' , hence also along ω . Therefore

$$\|W_n - 1\|_2 \geq \|(-1) - 1\|_2 - \|W_n - (-1)\|_2 \geq 2 - \|W_n + 1\| \xrightarrow{n \rightarrow \omega} 2,$$

contradicting (8). This contradiction shows $\Theta(w) = 1$, proving part (2).

For the final assertions of Theorem A: an MF embedding of E would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element w to 1. If $C_{\text{red}}^*(E)$ were an MF algebra, a unital C^* -embedding $C_{\text{red}}^*(E) \hookrightarrow \mathcal{Q}$ composed with the canonical map $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$, would exhibit an injective corona representation of E : the map $g \mapsto \lambda_g$ is injective, although the quotient map $C_{\text{max}}^*(E) \rightarrow C_{\text{red}}^*(E)$ need not be faithful. The canonical map $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$ is also injective, since its composition with the regular representation is $g \mapsto \lambda_g$; this treats $C_{\text{max}}^*(E)$. $\square$

Remark 8.1. The proof shows more than non-injectivity of each individual Θ : the fixed element w lies in the kernel of *every* corona representation of E , so every corona representation factors through $E/\langle w \rangle$ (note $\langle w \rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, w is a nontrivial element of the “MF radical” of E .

Remark 8.2. No lower bound on the corner ranks r_n , or on their density in d_n , is used anywhere: Lemma 6.3 renormalizes the trace so that all subsequent estimates are relative to the corner. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 10).

9. THE CYCLIC-BASE COMPARISON

We prove Theorem C. Recall the setting: $G_0 = \text{BS}(1, 2)$ with base $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$ and stable letter t satisfying $t\gamma_0 t^{-1} = \gamma_0^2$; $X_0 = G_0/\Gamma_0$; $E_0 = \text{Cilamp}(X_0) \rtimes G_0$ with central sign $z_0 = -1 \in \text{Cilamp}(X_0)$ as in Construction 4.1.

Proof of Theorem C. $z_0 \neq 1$: immediate from Construction 4.1, since $\text{Cilamp}(X_0) \leq \text{Cl}(X_0)^\times$ and $\text{Cl}(X_0) \neq 0$.

Amenability: every finitely generated subgroup of $\text{Cilamp}(X_0)$ is contained in the subgroup generated by ± 1 and finitely many c_x , which is a finite 2-group (it has order at most 2^{k+1} for k sites, since every element is $\pm a$ reduced product of distinct generators). Hence $\text{Cilamp}(X_0)$ is locally finite, so amenable. The group $G_0 = \text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ is metabelian, so amenable. As an extension of the amenable G_0 by the amenable $\text{Cilamp}(X_0)$, the group E_0 is amenable.

MF: by Facts 2.3 and 2.4, the amenable group E_0 is MF, and an MF embedding $E_0 \hookrightarrow \mathcal{U}(\mathcal{Q})$ is injective, so the image of z_0 is a nontrivial (self-adjoint) unitary in the corona: the central sign survives.

The quotient from the literal presentation: assign $\gamma_0 \mapsto \gamma_0$, $t \mapsto t$, and $c \mapsto c_o$, where $o = \Gamma_0 \in X_0$. The point o is fixed by Γ_0 , so $[c_o, \gamma_0] = 1$. Also $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$. Moreover $\gamma_0 t\Gamma_0 \neq t\Gamma_0$ in G_0 : otherwise $t^{-1}\gamma_0 t \in \Gamma_0$; in the faithful affine model $\text{BS}(1, 2) = \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ with $\gamma_0 = (1, 0)$ and $t = (0, 1)$ acting by multiplication by 2, we compute $t^{-1}\gamma_0 t = (\frac{1}{2}, 0) \notin \Gamma_0 = \mathbb{Z} \times \{0\}$. Consequently

$$w_{\text{BS}} \mapsto [c_{t\Gamma_0}, c_{\gamma_0 t\Gamma_0}] = z_0.$$

The image is central of order two, so every relator of E_{BS} is respected. The resulting homomorphism is surjective: γ_0, t generate G_0 , the conjugates of c_o give all c_x because G_0 acts transitively on X_0 , and their commutators contain z_0 . Composing this quotient with the MF embedding of E_0 gives the asserted corona representation of E_{BS} .

The finite-dimensional obstruction persists: E_0 contains the pattern of Theorem B with $H = E_0$, $\Gamma = \Gamma_0$, $t = t$, $c = c_o$ ($o = \Gamma_0 \in X_0$, fixed by Γ_0), and $a = \gamma_0 \in \Gamma_0$. We have $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$ and $[c_o, \Gamma_0] = 1$.

Thus Theorem B says every finite-dimensional representation of E_0 , over every field, kills z_0 . Applying the same theorem directly to the displayed presentation of E_{BS} shows that every finite-dimensional representation kills w_{BS} . $\square$

Remark 9.1. Theorem C shows that the argument cannot be generalized merely by deleting its fixed-space rigidity input. The elementary Theorem B is genuinely weaker than the corona conclusion: E_0 has MF approximations even though its finite-dimensional representations all kill the sign.

10. LIMITATIONS: HILBERT–SCHMIDT METRICS AND HYPERLINEARITY

The proof of Theorem A does not extend to tracial ultraproducts, for a structural reason worth recording.

Suppose one replaces the norm corona $\mathcal{Q}$ by a tracial ultraproduct $\mathcal{M} = \prod^\omega (M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, with defects measured in $\|\cdot\|_2$ rather than operator norm, and asks whether a hyperlinear-style embedding of E must kill w . Two steps break.

First, Lemma 7.1 fails: Hilbert–Schmidt-multiplicative lifts need not be operator-norm multiplicative, so the adjoint maps $\text{Ad } U_{g,n}$ need not assemble into a homomorphism into any norm corona. One can instead pass to the conjugation representation on $L^2(\mathcal{M})$, which is an honest representation of E .

The tracial ultraproduct $\mathcal{M}$ itself is a finite von Neumann algebra. Second, and decisively, Lemma 6.2 is unavailable only after the argument passes to its conjugation action: $B(L^2(\mathcal{M}))$ is properly infinite, and the Kazhdan projections onto the fixed spaces of Γ and of $t\Gamma t^{-1}$, though unitarily equivalent via the shift and nested by the same argument as in Lemma 7.2, may satisfy a *proper* inclusion $P < \beta(t)P\beta(t)^*$: exactly the unilateral-shift phenomenon. There is no normalized finite trace on $B(L^2(\mathcal{M}))$ with which to compare these projections. In particular the present paper proves nothing about hyperlinearity or Connes embeddability of E , and nothing about the sofic-versus-hyperlinear problem.

The dividing line in the present argument is thus: *operator-norm matrix models form a stably finite world, in which property (T) compression is rigid; tracial matrix models form a properly infinite world at the level of their conjugation actions, in which the same projection comparison need not collapse to equality.*

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REFERENCES

- [AW] C. A. Akemann and M. E. Walter, *Unbounded negative definite functions*, Canad. J. Math. **33** (1981), no. 4, 862–871.

- [BDL] G. Bachner, S. Dogon, and A. Lubotzky, *On L^1 -approximation of groups*, J. Algebra **702** (2026), 235–243, to appear, [doi:10.1016/j.jalgebra.2026.04.041](https://doi.org/10.1016/j.jalgebra.2026.04.041).
- [BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's property (T)*, New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008.
- [BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^* -algebras*, Math. Ann. **307** (1997), no. 3, 343–380.
- [BO] N. P. Brown and N. Ozawa, *C^* -algebras and finite-dimensional approximations*, Graduate Studies in Mathematics, vol. 88, American Mathematical Society, Providence, RI, 2008.
- [CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^* -algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152.
- [Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat K -theory*, Adv. Math. **384** (2021), Paper No. 107722.
- [CRW] M. D. E. Conder, E. F. Robertson, and P. D. Williams, *Presentations for 3-dimensional special linear groups over integer rings*, Proc. Amer. Math. Soc. **115** (1992), no. 1, 19–26, [doi:10.2307/2159559](https://doi.org/10.2307/2159559).
- [DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and non-approximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, [doi:10.1017/fms.2020.5](https://doi.org/10.1017/fms.2020.5).
- [KT] G. Kun and A. Thom, *Nonsofic wreath products of residually finite groups*, preprint, 2026, [arXiv:2608.06222](https://arxiv.org/abs/2608.06222).
- [OAI] OpenAI, *A counterexample to the soficity conjecture*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 77–93, 2026, <https://cdn.openai.com/pdf/ten-proofs-oai.pdf>.
- [Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, [arXiv:2603.13564](https://arxiv.org/abs/2603.13564).
- [TWW] A. Tikuisis, S. White, and W. Winter, *Quasidiagonality of nuclear C^* -algebras*, Ann. of Math. (2) **185** (2017), no. 1, 229–284.